Joseph Chifuka Ngulube

Microbiologist | National AMR Laboratory Specialist | Laboratory Quality Management

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Professional Summary

Microbiologist and biomedical scientist with 15 years in clinical and public health microbiology in Zambia. Since October 2026, serves as Zambia's National AMR Laboratory Specialist with IQLS, supporting the national AMR surveillance plan and the National Coordinating Center.

Previously supervised microbiology at the CIDRZ Central Laboratory and provided technical oversight to a 12-laboratory One Health AMR surveillance network (7 human, 5 animal health) under the Fleming Fund. Skilled in bacteriology, antimicrobial susceptibility testing to CLSI standards, ISO 15189 quality management, method validation and outbreak confirmation. An experienced trainer and mentor with four peer-reviewed publications, registered with the Health Professions Council of Zambia since 2011.

Highlights and Recognition

- Set up the cholera diagnosis centre at Mporokoso District Hospital during the 2018 outbreak and confirmed the outbreak at Shibuyunji fishing camp.
- Member of the Africa CDC AVoHC-SURGE deployable health workforce (2026).
- "Most Innovative", CIDRZ Labour Day Award, 2023.
- Best Medical Laboratory Technologist of the Year, UTH Department of Pathology and Microbiology, 2015.

Professional Experience

National AMR Laboratory Specialist (Independent Consultant)

IQLS, Zambia | Oct 2026 to Present (part-time through Mar 2027)

- Support Zambia's national AMR surveillance plan and the National Coordinating Center.
- Strengthen quality-assured AMR testing in national reference and high-volume laboratories, including ESBL and MRSA detection.
- Mentor laboratory staff across participating sites to build AMR testing capacity and quality practice.
- Support continuity of reagent and consumable supply for AMR testing programmes.
- Support AMR data integration and reporting through DISIS/DHIS2 and One Health dashboards.

Trainee Diagnostics and Applications Sales Specialist

Sterelin Medical and Diagnostics, Lusaka | Sep 2026 to Oct 2026

Microbiology Laboratory Supervisor

CIDRZ Central Laboratory, Lusaka | Apr 2024 to Sep 2026

- Led daily operations of a high-throughput microbiology laboratory serving clinical diagnostics, research studies and public health surveillance.

- Supervised and mentored Biomedical Scientists, Technologists, interns and students, and set up structured onboarding and continuous training.
- Led validation, verification and implementation of new assays, reagent lots and testing procedures.
- Ran QC/QA programmes aligned with GCLP, GLP and ISO 15189, and led root-cause investigations and CAPA for nonconformities.
- Coordinated installation, calibration, preventive maintenance and troubleshooting of laboratory equipment.
- Coordinated compliant shipment of M. tuberculosis isolates and biological specimens under national and international transport rules.
- Supported internal audits, external quality assessment, accreditation activities and grant applications.

Microbiologist

CIDRZ Fleming Fund, Lusaka | Mar 2020 to Mar 2024

- Provided technical oversight to a 12-laboratory One Health AMR surveillance network (7 human, 5 animal health), improving AMR data quality.
- Developed, validated and rolled out standardised SOPs and bench aids for bacteriology and AST across surveillance sites.
- Delivered on-site technical assistance, mentorship and hands-on training in bacteriology, AST and quality management.
- Performed and supervised complex bacteriology testing on clinical and surveillance specimens, supporting outbreak response.
- Coordinated calibration, maintenance and QC for BD BACTEC, BD Phoenix and VITEK 2/MS systems.
- Wrote annual workplans and donor reports and contributed to national AMR data analysis informing policy and stewardship.

Medical Laboratory Technologist

Ministry of Health / University Teaching Hospitals, Lusaka | Oct 2012 to Dec 2019

- Performed routine and specialised bacteriology and mycology testing on diverse clinical specimens at Zambia's largest tertiary referral hospital.
- Isolated and identified bacteria and fungi and performed AST using standardised methods.
- Validated, interpreted and reported microbiology and AST results, communicating critical findings promptly.
- Supported diagnosis of infectious disease outbreaks and complex clinical cases.
- Trained and supervised new staff, students and interns on procedures, equipment and biosafety.

Medical Laboratory Technologist / Malaria Microscopist

Lusaka Provincial Health Office / CIDRZ IeDEA Malaria, Chongwe | Jun 2011 to Sep 2012

- Established laboratory services at Chinyunyu Rural Health Centre as its first laboratory professional, with workflows for testing, documentation and referral.
- Diagnosed malaria by microscopy and RDTs, including species identification and parasite quantification.
- Prepared Ziehl-Neelsen smears for TB and referred specimens for confirmatory testing; performed phlebotomy for ART and antenatal programmes.
- Trained and supervised research assistants in malaria microscopy, staining and RDT procedures.

Education

Bachelor of Science in Biomedical Sciences

University of Zambia, Lusaka | 2019

Diploma in Biomedical Sciences

Ndola College of Biomedical Sciences, Ndola | 2010

Grade 12 Certificate

Kantanshi Secondary School, Mufulira | 2005

Certifications and Registration

- BSAFE Course, United Nations Department of Safety and Security, 2026
- CITI Good Clinical Practice (GCP), University of Alabama at Birmingham, 2026
- Mentors Trainer of Trainers, Ministry of Health / UK Health Security Agency, 2025
- Enteric Pathogens Trainer, Ministry of Health / UK Health Security Agency, 2024
- Leadership and Management in Health, University of Washington, 2023
- Commodity Logistics Management, Medical Stores / JICA, 2010
- Full Registration, Health Professions Council of Zambia (HPCZ), since 2011, with a 2026 practising certificate

Core Skills

- **Microbiology:** bacterial culture and isolation, aerobic and anaerobic bacteriology, blood culture, fungal culture and identification, antimicrobial susceptibility testing (CLSI), ESBL and MRSA detection, culture media preparation and QC
- **Laboratory techniques:** Gram and Ziehl-Neelsen staining, wet preparations, malaria microscopy and parasite quantification, RDTs, phlebotomy, isolate preservation, specimen shipment and chain of custody
- **Instruments:** BD BACTEC, BD Phoenix, VITEK 2 Compact, VITEK MS (MALDI-TOF), Class II biosafety cabinets, light and fluorescence microscopy, autoclaves, incubators, cold chain
- **Quality management:** ISO 15189, GCLP, GLP, IQC, EQA, method validation and verification, root-cause analysis, CAPA, nonconformity management, document control, accreditation readiness
- **Biosafety:** BSL-2 operations, biosafety risk assessment, infection prevention and control, biological and chemical waste management, biosecurity
- **AMR surveillance and public health:** One Health AMR surveillance, Fleming Fund programme implementation, outbreak investigation and confirmation, site assessments, laboratory capacity strengthening
- **Data and reporting:** WHONET, GLASS reporting, DHIS2, LIS/LIMS, monitoring and evaluation, technical and donor report writing, Microsoft 365
- **Leadership:** laboratory supervision, staff training and performance management, mentorship, inventory and procurement planning, stakeholder engagement

Publications

- Hatyoka et al. (2022). Antimicrobial Susceptibility Patterns of E. coli and Shigella Isolated from Stool Samples from Adults and Children in Zambia.

- Mushabati et al. (2021). Bacterial Contamination of Mobile Phones of Healthcare Workers at the University Teaching Hospital, Lusaka, Zambia.
- Mwape et al. (2020). Characterization of Vibrio cholerae Isolates from the 2009, 2010 and 2016 Cholera Outbreaks in Lusaka Province, Zambia.
- Mutengo et al. (2017). A Case Report: Primary Amoebic Meningoencephalitis in a Young Zambian Adult.

Professional Affiliations

- Member, Africa CDC AVoHC-SURGE (African Volunteers Health Corps) deployable workforce, 2026
- Technical Working Group on Research and Surveillance, Zambia, 2023
- Biomedical Society of Zambia, 2014

Languages

English (full professional proficiency)

References

Available on request.